

FINAL EXPLANATION: BIOLOGY GRADE 11

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: BIOLOGY GRADE 11: EXCRETION AND HOMEOSTASIS IN ANIMALS

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Welcome to your Final Explanation. Think back to the hot afternoon in a Kenyan community: a dog pants with its tongue out, a lizard sits still on a warm rock, goats drink water yet make concentrated urine, and freshwater fish keep excreting large amounts of dilute urine. Use everything you observed and learned in Lessons 1 to 7 to answer the questions below. Write your answers in the space under each question. Use evidence from your kidney dissection, your charts, your 'Follow the Fluid' table, your feedback flowcharts and your class model. Use correct terms such as nitrogenous waste, ultrafiltration, reabsorption, ADH, negative feedback, endotherm and ectotherm. Where a question asks for a claim, evidence and reasoning, state your claim first, support it with evidence, then explain why the evidence supports it. Your last task is to plan the annotated poster that will be your final product.

Part 1: Getting rid of nitrogenous wastes in different habitats

Animals make toxic nitrogenous wastes when they break down proteins. (a) Name the three nitrogenous wastes you studied and say which animals you linked to each one: freshwater fish, marine fish, insects, birds and mammals. (b) Explain how the type of waste an animal makes is linked to how much water is available to it. Use the freshwater fish and the insects and birds in your explanation. (c) Name the excretory structures used by freshwater fish, marine fish, insects and birds.

(a) The three wastes are ammonia, urea and uric acid. Freshwater fish excrete ammonia. Marine fish and mammals make urea. Insects and birds make uric acid. (b) The type of waste is linked to water availability. Freshwater fish have plenty of water available, and they excrete toxic ammonia in dilute urine. Insects and birds must save water, and they excrete less toxic uric acid as a semi-solid paste. (c) Freshwater fish and marine fish use gills and kidneys. Insects use Malpighian tubules. Birds use kidneys. In each case the structure removes nitrogenous waste in a way that suits the water available to the animal.

Part 2: How the mammalian kidney makes urine

Think about the goat kidney you dissected and the animations of the nephron. (a)

(a) The cortex was the dark outer region with granular dots; it contains many nephrons where blood is filtered. The medulla was paler and striped; it contains parts of the nephrons and collecting ducts that help concentrate urine. The pelvis was a hollow space that collects urine and passes it to the ureter. Blood vessels (renal artery and vein) entered and left at the hilum, showing a large

Describe what you saw in the cortex, medulla and pelvis, and say what each region does. (b) Using your 'Follow the Fluid' table, explain the steps by which blood becomes urine. Include ultrafiltration, selective reabsorption and secretion. (c) Explain why glucose is found in blood and filtrate but not in urine, while urea is more concentrated in urine. (d) Which hormone controls how much water is reabsorbed, and where in the nephron does it act?	blood supply. The nephron is the functional unit of the kidney. (b) In the glomerulus, blood is under high pressure, so small molecules (water, glucose, urea, salts) are forced into Bowman's capsule. This is ultrafiltration. Blood cells and proteins are too large and stay in the blood. As the filtrate moves along the tubule, useful substances are selectively reabsorbed into the blood, and extra wastes and ions are secreted into the tubule. The remaining fluid is urine. It passes through the collecting duct to the pelvis, ureter and bladder. (c) All glucose is reabsorbed in the proximal tubule, so none is left in urine. Water is reabsorbed in the loop of Henle and collecting duct, so the volume of fluid falls and urea becomes more concentrated in the urine. (d) ADH controls how much water is reabsorbed. It acts on the collecting duct, so urine can be concentrated or dilute.
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Part 3: The liver and skin, and the goat and the freshwater fish	
(a) Where does the urea in a goat's urine come from, and how does it reach the urine? (b) State one other job of the liver and one job of the skin in excretion. (c) Explain why a goat that has lost water on a hot day makes small amounts of concentrated urine, while a freshwater fish makes large amounts of dilute urine. Write this as a claim, evidence and reasoning explanation, using what you learned about the kidney for the goat and about water movement for the fish.	(a) The urea starts in the liver, where excess amino acids are deaminated to make urea. Blood carries the urea to the kidney, where it is filtered at the glomerulus and removed in the urine. (b) The liver also breaks down haemoglobin to make bile pigments and processes toxic substances. The skin excretes sweat containing water, salts and a little urea, and sweat helps cool the body. (c) Claim: The goat conserves water by making concentrated urine, while the freshwater fish gets rid of extra water by making dilute urine. Evidence: In our Follow the Fluid table, urine became concentrated when water was reabsorbed in the loop of Henle and collecting duct. Freshwater fish gain water by osmosis and excrete large amounts of dilute urine. Reasoning: A goat that loses water on a hot day releases more ADH. ADH makes the collecting ducts reabsorb more water back into the blood, so little water is left in the urine, which is concentrated. The freshwater fish takes in water by osmosis from its surroundings, so it must remove the excess and does not need to save water, so its urine is dilute and plentiful. Both are controlling water balance to keep the internal environment stable.

Part 4: Negative feedback and the dog and the lizard	
(a) Explain what negative feedback means, using a set point. (b) Draw or describe a feedback loop for either blood glucose after a meal of ugali and sukuma wiki, or body temperature on a hot day. Name	(a) Negative feedback detects a change away from a set point and triggers a response that reverses the change, bringing the body back to normal. (b) Temperature on a hot day: thermoreceptors detect that the body is too warm. The hypothalamus is the control centre. Effectors respond: sweat glands release sweat, skin arterioles widen (vasodilation) so more heat is lost, and panting increases. Body temperature falls back to the set point and the response slows. (Blood glucose alternative: after a meal glucose rises, the pancreas releases insulin, glucose levels fall; if glucose falls too low, glucagon is released to raise it.) (c) The dog is an endotherm. It pants so that water evaporates from its tongue and airways and carries heat away, and its skin blood vessels dilate to lose heat, so its body temperature stays steady. The lizard is an ectotherm. It cannot control its temperature internally, so it

the detector, the control centre and the effectors or hormones. (c) Explain why the dog pants and why the lizard sits still on the warm rock. In your answer, use the words endotherm and ectotherm. (d) Name two behaviours, other than basking and seeking shade, that animals use to survive extreme cold or dry conditions.	relies on behaviour: it gains heat by basking on the warm rock and moves into shade or a crevice when it gets too hot. Both are using homeostasis to survive the heat. (d) Any two of: burrowing, hibernation, aestivation. These help animals survive prolonged cold or hot, dry periods.
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Part 5: Your final poster and why homeostasis matters

Your final product is an annotated poster model explaining how a mammal (a goat or a human) regulates its internal environment and excretes waste. (a) List at least five labelled parts or processes your poster must include and say what each shows. (b) Explain how your poster model is different from your first model in Lesson 1. (c) Explain why excretion and homeostasis are essential for the survival of the cells and health of all living organisms.	(a) Possible poster parts: 1) the kidney with cortex, medulla and pelvis, showing where filtering and urine concentration happen; 2) a nephron showing ultrafiltration at the glomerulus, selective reabsorption and secretion; 3) the liver, showing deamination to make urea and bile production; 4) the skin, showing sweat glands and the loss of water, salts and a little urea; 5) feedback loops: ADH for water balance, insulin and glucagon for blood glucose, and a temperature loop using the hypothalamus, vasodilation, vasoconstriction, sweating and shivering. Each part should have a label and a one-sentence note on its function, linked to the hot afternoon puzzle. (b) My Lesson 1 model only showed that animals have different wastes and could not explain panting, basking or concentrated urine. My final model adds the kidney structure, nephron processes, the liver and skin, and negative feedback control, so it explains how the goat makes concentrated urine, why the dog pants, and why the lizard basks. (c) Cells only work properly in a stable internal environment with steady temperature, water, salts and blood sugar. Toxic wastes such as ammonia and urea would build up and damage cells if they were not removed. Excretion and homeostasis prevent this, so animals in different habitats survive and stay healthy despite changes in the environment.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explaining excretion and nitrogenous wastes	Correctly names ammonia, urea and uric acid, matches them to animals and links each to water availability with clear reasoning.	Names the wastes and matches most to the correct animals, with a basic link to water availability.	Names some wastes but mixes up animals or does not link waste type to water.
Kidney structure and urine formation	Accurately describes cortex, medulla and pelvis and explains ultrafiltration, selective reabsorption, secretion and the role of ADH in the correct order, using	Describes kidney regions and most urine formation steps, with minor gaps in sequence or terms.	Names some parts or steps but with inaccuracies or missing links between structure and function.

	evidence from the Follow the Fluid table.		
Negative feedback and thermoregulation	Explains set point, detector, control centre and effectors correctly for at least one loop, and clearly contrasts the dog (endotherm) and lizard (ectotherm).	Gives a mostly correct feedback loop and a reasonable explanation of the dog and lizard.	Describes a response but does not show the feedback idea or confuses endotherm and ectotherm.
Use of evidence and claim-evidence-reasoning	Makes clear claims supported by specific evidence from the lessons and reasoning that connects the evidence to the hot afternoon phenomenon.	Makes claims with some evidence, but the reasoning is partly incomplete.	Gives statements with little or no evidence or reasoning.
Poster model and communication	Poster plan includes at least five accurate, labelled parts with functions, shows links between kidney, liver, skin and feedback control, and explains why homeostasis matters for cells.	Poster plan includes about four accurate labelled parts with some links between them.	Poster plan has few or unclear labels and little connection between parts.

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